

Solo E-Commerce - Database ER Diagram

This document provides a visual representation of the database schema using Mermaid diagrams.

Complete ER Diagram

erDiagram

%% ===== PUBLIC SCHEMA - USER & AUTH =====

User ||--o{ Address : "has many"
User ||--o| Cart : "has one"
User ||--o{ Order : "has many"
User ||--o{ RefreshToken : "has many"
User ||--o| LoyaltyWallet : "has one"
User ||--o{ SavedPaymentMethod : "has many"
User ||--o{ AnalyticsEvent : "generates"

User {
 uuid id PK
 string email UK
 string passwordHash
 string firstName
 string lastName
 string phone
 UserRole role
 boolean isActive
 boolean emailVerified
 datetime createdAt
 datetime updatedAt
}

RefreshToken {
 uuid id PK
 string token UK
 uuid userId FK
 datetime expiresAt
 boolean isRevoked
}

LoyaltyWallet ||--o{ LoyaltyTransaction : "has many"

LoyaltyWallet {
 uuid id PK
 uuid userId FK
 decimal balanceAed
 decimal totalEarnedAed
 decimal totalRedeemedAed
}

LoyaltyTransaction {
 uuid id PK
 uuid walletId FK
 LoyaltyTransactionType type
 decimal amountAed
 string description
 uuid orderId

```

}

Address {
    uuid id PK
    uuid userId FK
    string label
    string firstName
    string lastName
    string addressLine1
    string city
    string postalCode
    boolean isDefault
}

SavedPaymentMethod {
    uuid id PK
    uuid userId FK
    string provider
    string brand
    string last4
    int expMonth
    int expYear
}

%% ===== PUBLIC SCHEMA - CATALOG =====

Department ||--o{ Category : "contains"

Department {
    uuid id PK
    string name UK
    string slug UK
    string description
    int sortOrder
    boolean isActive
}

Category ||--o{ Category : "parent-child"

Category {
    uuid id PK
    uuid departmentId FK
    uuid parentId FK
    string name
    string slug UK
    string description
    string image
    int sortOrder
}

```

```

Brand {
    uuid id PK
    string name UK
    string slug UK
    string description
    string logo
    int sortOrder
}

ProductOverride {
    uuid id PK
    string inventorySku UK
    int inventoryId
    boolean isFeatured
    boolean isNew
    boolean isBestSeller
    decimal customPrice
    decimal customSalePrice
    string metaTitle
}

%% ===== PUBLIC SCHEMA - CART & ORDERS =====

Cart ||--o{ CartItem : "contains"

Cart {
    uuid id PK
    uuid userId FK
    datetime createdAt
}

CartItem {
    uuid id PK
    uuid cartId FK
    CartItemType type
    int productId
    uuid packageId FK
    int quantity
}

Order ||--o{ OrderItem : "contains"
Order ||--o{ OrderStatusHistory : "has"
Order }o--|| Address : "shipping"
Order }o--|| Address : "billing"

Order {
    uuid id PK
    string orderNumber UK
    uuid userId FK
    OrderStatus status
}

```

```

    PaymentStatus paymentStatus
    uuid shippingAddressId FK
    uuid billingAddressId FK
    ShippingMethod shippingMethod
    decimal subtotal
    decimal discount
    decimal tax
    decimal total
    string promoCode
    PaymentMethod paymentMethod
}

```

```

OrderItem {
    uuid id PK
    uuid orderId FK
    CartItemType type
    int productId
    string name
    string sku
    int quantity
    decimal price
    decimal subtotal
}

```

```

OrderStatusHistory {
    uuid id PK
    uuid orderId FK
    OrderStatus status
    string notes
    string createdBy
}

```

```

PromoCode {
    uuid id PK
    string code UK
    PromoType type
    decimal value
    decimal minOrderAmount
    int usageLimit
    int usageCount
    boolean isActive
    datetime startsAt
    datetime expiresAt
}

```

```

%% ===== PUBLIC SCHEMA - CMS =====

```

```

Banner ||--o{ LandingPage : "hero for"

```

```

Banner {

```

```

        uuid id PK
        BannerPlacement placement
        string title
        string subtitle
        string ctaText
        string ctaUrl
        string imageDesktopUrl
        int displayOrder
        boolean isActive
    }

    LandingPage ||--o{ LandingSection : "has"

    LandingPage {
        uuid id PK
        string slug UK
        string title
        uuid heroBannerId FK
        string seoTitle
        text seoDescription
        boolean isActive
    }

    LandingSection {
        uuid id PK
        uuid landingPageId FK
        LandingSectionType type
        string title
        text data
        text config
        int displayOrder
        boolean isActive
    }

    %% ===== PUBLIC SCHEMA - NAVIGATION =====

    NavigationMenu ||--o{ NavigationMenuItem : "contains"

    NavigationMenu {
        uuid id PK
        string key UK
        string name
        boolean isActive
    }

    NavigationMenuItem ||--o{ NavigationMenuItem : "children"

    NavigationMenuItem {
        uuid id PK
        uuid menuId FK

```

```

        uuid parentId FK
        string label
        string url
        string icon
        string badge
        int sortOrder
    }

%% ===== PUBLIC SCHEMA - BLOG =====

BlogCategory ||--o{ BlogPost : "contains"

BlogCategory {
    uuid id PK
    string name UK
    string slug UK
    string description
    int sortOrder
}

BlogPost ||--o{ BlogPostTag : "has"
BlogTag ||--o{ BlogPostTag : "tagged in"

BlogPost {
    uuid id PK
    uuid categoryId FK
    string title
    string slug UK
    text excerpt
    text content
    string featuredImage
    string author
    boolean isFeatured
    boolean isActive
    datetime publishedAt
}

BlogTag {
    uuid id PK
    string name UK
    string slug UK
}

BlogPostTag {
    uuid id PK
    uuid postId FK
    uuid tagId FK
}

%% ===== PUBLIC SCHEMA - HOME CMS =====

```

```

HomePageConfig ||--o{ HomePageSection : "has"

HomePageConfig {
    cuid id PK
    string key UK
}

HomePageSection {
    cuid id PK
    string homePageId FK
    HomeSectionType type
    string title
    int position
    boolean isEnabled
    json config
}

CategoryLandingPageConfig ||--o{ CategoryLandingSection : "has"

CategoryLandingPageConfig {
    cuid id PK
    string categoryId UK
    string heroTitle
    string heroImageUrl
    boolean isHeroEnabled
}

CategoryLandingSection {
    cuid id PK
    string landingId FK
    CategoryLandingSectionType type
    string title
    int position
    json config
}

%% ===== PUBLIC SCHEMA - ANALYTICS =====

AnalyticsEvent {
    uuid id PK
    AnalyticsEventType type
    uuid userId FK
    int productId
    text metadata
    string sessionId
    datetime createdAt
}

SavedSearchTerm {

```



```
uuid id PK
string term
int resultCount
uuid userId
string sessionId
}
```

Inventory Schema ER Diagram

erDiagram

```
%% ===== INVENTORY SCHEMA =====

InvCountry ||--o{ InvProduct : "origin of"
InvBrand ||--o{ InvProduct : "makes"
InvDesigner ||--o{ InvProduct : "designs"
InvCategory ||--o{ InvSubcategory : "contains"
InvCategory ||--o{ InvProduct : "categorizes"
InvSubcategory ||--o{ InvProduct : "subcategorizes"

InvProduct ||--o| InvProductDimension : "has"
InvProduct ||--o| InvProductPackaging : "has"
InvProduct ||--o| InvProductPricing : "has"
InvProduct ||--o{ InvProductImage : "has many"
InvProduct ||--o{ InvProductSpecification : "has many"
InvProduct ||--o{ InvInventoryTransaction : "tracks"

InvCountry {
    int id PK
    string countryCode UK
    string countryName
}

InvBrand {
    int id PK
    string brandName UK
    text description
    string website
    boolean isActive
}

InvDesigner {
    int id PK
    string designerName UK
    text bio
    boolean isActive
}

InvCategory {
    int id PK
    string categoryName UK
    text description
    int displayOrder
    boolean isActive
}

InvSubcategory {
    int id PK
    int categoryId FK
    string subcategoryName
```

```

        text description
        int displayOrder
        boolean isActive
    }

    InvProduct {
        int id PK
        string sku UK
        string productName
        text description
        text shortDescription
        text fullDescription
        json highlights
        json galleryImageUrls
        json specs
        string urlSlug
        int categoryId FK
        int subcategoryId FK
        int brandId FK
        int designerId FK
        int countryId FK
        string material
        string colour
        boolean isActive
        boolean isFeatured
        boolean isNew
        boolean isBestSeller
    }

    InvProductDimension {
        int id PK
        int productId FK
        decimal functionalWidthCm
        decimal functionalHeightCm
        decimal functionalDepthCm
        decimal packedWeightKg
        decimal productWeightKg
    }

    InvProductPackaging {
        int id PK
        int productId FK
        string packagingType
        int colliSize
        decimal colliWeightKg
        decimal colliLengthCm
        decimal colliWidthCm
    }

    InvProductPricing {

```

```

    int id PK
    int productId FK
    decimal rrpAedExclVat
    decimal priceInclVat
    decimal listedPriceVat
    string currency
    decimal vatRate
    boolean isCurrent
    date effectiveFrom
    date effectiveTo
}

InvProductImage {
    int id PK
    int productId FK
    string imageUrl
    string imageType
    string altText
    int displayOrder
    boolean isPrimary
}

InvProductSpecification {
    int id PK
    int productId FK
    string specKey
    text specValue
    string specUnit
    int displayOrder
}

InvInventoryTransaction {
    int id PK
    int productId FK
    string transactionType
    int quantity
    int quantityBefore
    int quantityAfter
    string reference
    text notes
}

```

Relationship Summary

Public Schema Relationships

Parent	Child	Type	Description
User	Address	1:N	User has many addresses
User	Cart	1:1	User has one cart
User	Order	1:N	User has many orders
User	LoyaltyWallet	1:1	User has one wallet
User	RefreshToken	1:N	User has many tokens
Department	Category	1:N	Department has categories
Category	Category	1:N	Self-referential hierarchy
Cart	CartItem	1:N	Cart has many items
Order	OrderItem	1:N	Order has many items
Order	Address	N:1	Order has shipping/billing address
LandingPage	LandingSection	1:N	Page has many sections
NavigationMenu	NavigationMenuItem	1:N	Menu has many items
BlogCategory	BlogPost	1:N	Category has many posts
BlogPost	BlogPostTag	1:N	Post has many tags

Inventory Schema Relationships

Parent	Child	Type	Description
InvCategory	InvSubcategory	1:N	Category has subcategories
InvCategory	InvProduct	1:N	Category has products
InvSubcategory	InvProduct	1:N	Subcategory has products
InvBrand	InvProduct	1:N	Brand has products
InvProduct	InvProductPricing	1:1	Product has pricing
InvProduct	InvProductDimension	1:1	Product has dimensions
InvProduct	InvProductImage	1:N	Product has images
InvProduct	InvProductSpecification	1:N	Product has specs

Cross-Schema References

The public schema references inventory schema products through:

- `CartItem.productId` → `InvProduct.id`
- `OrderItem.productId` → `InvProduct.id`
- `ProductOverride.inventoryId` → `InvProduct.id`
- `ProductOverride.inventorySku` → `InvProduct.sku`
- `AnalyticsEvent.productId` → `InvProduct.id`

These are **logical references** (not database foreign keys) handled in the application service layer.

Enum Types

Public Schema Enums

```

-- User Role
CREATE TYPE "UserRole" AS ENUM ('CUSTOMER', 'ADMIN', 'SUPER_ADMIN');

-- Order Status
CREATE TYPE "OrderStatus" AS ENUM (
    'PENDING', 'PAYMENT_PENDING', 'PAID', 'PROCESSING',
    'SHIPPED', 'DELIVERED', 'CANCELLED', 'REFUNDED'
);

-- Payment Status
CREATE TYPE "PaymentStatus" AS ENUM (
    'PENDING', 'AUTHORIZED', 'PAID', 'FAILED', 'REFUNDED'
);

-- Payment Method
CREATE TYPE "PaymentMethod" AS ENUM ('CREDIT_CARD', 'CASH_ON_DELIVERY');

-- Shipping Method
CREATE TYPE "ShippingMethod" AS ENUM ('STANDARD', 'EXPRESS', 'OVERNIGHT', 'PICKUP');

-- Banner Placement
CREATE TYPE "BannerPlacement" AS ENUM (
    'HOME_HERO', 'HOME_MID', 'HOME_SECONDARY',
    'CATEGORY_TOP', 'CATEGORY_MID', 'CATEGORY',
    'PRODUCT_SIDEBAR', 'CHECKOUT_TOP', 'PROMOTION'
);

-- Landing Section Type
CREATE TYPE "LandingSectionType" AS ENUM (
    'PRODUCT_GRID', 'CATEGORY_GRID', 'RICH_TEXT', 'IMAGE',
    'BANNER_CAROUSEL', 'HERO', 'CATEGORY_TILES', 'PRODUCT_CAROUSEL',
    'BRAND_STRIP', 'PROMO_BANNER', 'HERO_SLIDER', 'VALUE_PROPS_ROW',
    'SALE_STRIP_BANNER', 'BLOG_LATEST_GRID', 'TESTIMONIALS', ...
);

-- Analytics Event Type
CREATE TYPE "AnalyticsEventType" AS ENUM (
    'PAGE_VIEW', 'PRODUCT_VIEW', 'PRODUCT_SEARCH',
    'ADD_TO_CART', 'REMOVE_FROM_CART', 'CART_VIEW',
    'CHECKOUT_START', 'CHECKOUT_COMPLETE', 'ORDER_PLACED'
);

```

Model Statistics

Category	Count
Public Schema Models	~35

Category	Count
Inventory Schema Models	~10
Total Models	~45
Enum Types	~15
Foreign Key Relationships	~50
Unique Indexes	~30